2-Calabi-Yau Triangulated Categories + Zamolodchikov Periodicity 6/5/24

I. Intro to Cluster Algebras (Fomin + Zelevinsky, 2001) (book in 2016)

Def: given a quiver Q, $v \in Q$, can mutate at v to obtain a new quiver

(1) $\forall i \to v \to j$, add an arrow $i \to j$

(2) remove all 2-cycles

(3) flip all arrows adjacent to v

Def: we associate a variable x_v to each vertex $v \in Q$, and mutation at x_v replaces x_v with x_v' s.t. $x_v x_v' = \prod_{i \to v} x_i + \prod_{v \to j} x_j$

- we call the $\{x_v : v \in Q\}$ cluster variables
- the set of cluster variables in a given quiver form a cluster
- the cluster algebra A_Q over R (usually $\mathbb{C}$) is generated by all cluster variables across all mutations

II. Categorification

(not) Def: the idea of replacing things in cluster algebras with similar things in a category with additional structure

- usually f.d. modules over some f.d. k-algebra
- should have an associated quiver for which mutations commute

Note: cluster structures for categories originally defined in 2007

"cluster structures for 2-Calabi-Yau Categories + Unipotent Groups" by Buan, Iyama, Reiten, Scott

Why is categorification even useful???

(1) Can unexpectedly yield familiar cluster structures

(Pasha + Thomas, 2019): "Positroid Varieties + Cluster Algebras" categorification via preprojective algebras gives same cluster structure as coordinate ring of an open positroid variety

(2) (Caldero + Keller, 2006): if Q, Q' acyclic, mutation equiv, then Q can be transformed via mutations at sinks/sources into some $Q'' \cong Q'$. proved using cluster categories

(3) Zamolodchikov Periodicity: periodicity conjecture for pairs of Dynkin diagrams

Historical Note: "cluster category" was created in 2004 by Buan-Marsh-Reineke, Reiten, Todorov in "Tilting Theory + Cluster Combinatorics"

III. Zamolodchikov Periodicity

Def: Let Q be bipartite s.t. $\mu_\circ, \mu_\bullet$ just flips all arrows of Q.

the Y-system associated with Q is $Y_v(t)$ s.t. $\forall t \in \mathbb{Z}$,

$$Y_v(t+1)\,Y_v(t-1) = \prod_{u \to v} (1+Y_u(t)) \prod_{v \to w} (1+Y_w^{-1}(t))^{-1}$$ for v white, and

$$Y_v(t+1)\,Y_v(t-1) = \prod_{v \to u} (1+Y_u(t)) \prod_{w \to v} (1+Y_w^{-1}(t))^{-1}$$

Def: Q is Zamolodchikov periodic if the associated Y-system is periodic

The Periodicity Conjecture for pairs of Dynkin Diagrams (Keller, 2010)

Let Q be $\Delta \otimes \Delta'$ tensor product of two Dynkin diagrams.
Then the Y-system of Q is periodic $(Y_v(t+2N) = Y_v(t))$ with period dividing $h+h'$, the Coxeter numbers of Δ and Δ'.

Note: apparently significant in physics + dilogarithm identities
proof uses cluster categories, and the 2-Calabi-Yau ness of the cluster category. Zamolodchikov is Russian physicist, first conjectured for Δ simply laced, $\Delta' = A_1$ in study of thermodynamics

IV. the Cluster Category

"the Knitting Algorithm": Given an orientation of a simply laced Dynkin diagram Q, the repetition / Bratteli diagram $\mathbb{Z}Q$ is countable copies of Q so that arrows go right

(1) draw $\mathbb{Z} \times Q$

(2) $\forall\, \alpha: i \to j$ in Q, add arrows $(n,j) \to (n+1, i)$

(3) define $\tau: \mathbb{Z}Q \to \mathbb{Z}Q$ (maybe do an example w/ A_5)

$$(p,i) \mapsto (p-1, i) \quad \forall p \in \mathbb{Z}$$
$$(p,\alpha) \mapsto (p-1, \alpha) \quad \forall p \in \mathbb{Z}$$

the Path Category: Let k alg. closed, Q as above

- objects are vertices of Q
- morphisms are $\mathrm{Hom}(i,j) = \mathrm{Span}\{\text{paths } i \to j\}$
- composition = composition of paths
- $P_i = \{\text{all paths ending at } i\}$ is projective right module, as kQ an algebra
  these P_i generate category $\mathrm{mod}\, kQ$ of right kQ modules

the bounded derived Category $D_Q = D^b(\mathrm{mod}\, kQ)$

- objects are bounded complexes of right kQ-modules
- morphisms are morphisms of complexes
- D_Q has a natural shift functor Σ that shifts complexes to the right, making it into a triangulated category

Def: a category is triangulated if

(1) it is k-linear (additive, morphisms have k-v.s. structure)

(2) has a shift functor Σ

(3) has a class of "exact triangles" induced by s.e.s. of complexes satisfying certain axioms:

(TR1): $\forall\, X \xrightarrow{u} Y$, $\exists!$ cone Z (up to isom) s.t. $X \xrightarrow{u} Y \to Z \to \Sigma X$ exact Δ

(TR2): $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} \Sigma X$ exact Δ iff $Y \xrightarrow{v} Z \xrightarrow{w} \Sigma X \xrightarrow{\Sigma u} \Sigma Y$ exact Δ

Thm (Happel, 1987): $\exists$ bijection $\{\text{vertices} \in \mathbb{Z}Q\} \leftrightarrow \left\{\begin{matrix}\text{isom classes of indecomp.} \\ \text{objects in } D_Q\end{matrix}\right\}$

that lifts to an equivalence of categories $\mathrm{Mesh}(\mathbb{Z}Q) := \dfrac{k\mathbb{Z}Q}{\langle \alpha\alpha^* : \alpha \in Q\rangle} \rightleftarrows \mathrm{ind}(D_Q)$

s.t. $(1,i)$ corresponds to P_i

Note: each mesh
$$\begin{tikzcd} & u_1 \arrow[dr] & \\ \tau(v) \arrow[ur] \arrow[r] \arrow[dr] & u_2 \arrow[r] & v \\ & u_s \arrow[ur] & \end{tikzcd}$$
corresponds to an exact Δ, so $\underline{\dim}\, M_v + \underline{\dim}\, M_{\tau v} = \sum_{i=1}^{s} \underline{\dim}\, M_{u_i}$

$$M_{\tau v} \to \bigoplus_{i=1}^{s} M_{u_i} \to M_v \to \Sigma M_{\tau v}$$

called "Auslander-Reiten sequence" or "almost split sequence"

Example:

Consider the following orientation of A_5

$$1 \to 2 \to 3 \leftarrow 4 \to 5$$

since Grothendieck gps of mod kQ and D_Q are isomorphic

(1) Calculate dim vectors of the $P_i \Leftrightarrow$ dim vectors as kQ modules

since Q acyclic, there is at most one path from each j to i

$\underline{\dim}\, P_i = (v_1, v_2, v_3, v_4, v_5)$ where $v_j = \begin{cases} 1 & \text{if } \exists \text{ a path } j \to i \\ 0 & \text{otherwise} \end{cases}$

$\underline{\dim}\, P_1 = (1,0,0,0,0)$ 1

$\underline{\dim}\, P_2 = (1,1,0,0,0)$ 12

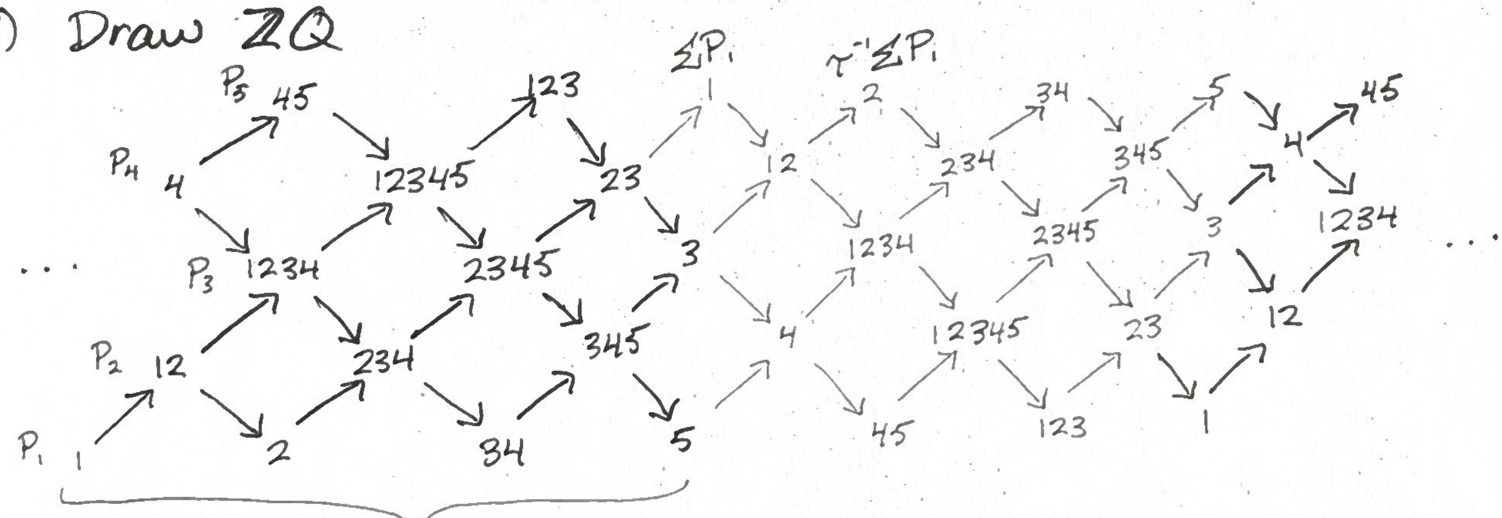
$\underline{\dim}\, P_3 = (1,1,1,1,0)$ 1234

$\underline{\dim}\, P_4 = (0,0,0,1,0)$ 4

$\underline{\dim}\, P_5 = (0,0,0,1,1)$ 45

(2) use $\underline{\dim}\, M_v + \underline{\dim}\, M_{\tau v} = \sum_{i=1}^{5} \underline{\dim}\, M_{u_i}$ to calculate all others

(3) Draw $\mathbb{Z}Q$



"Auslander-Reiten quiver"

Def: the cluster category $\mathcal{C}_Q = \dfrac{D_Q}{(\tau^{-1}\Sigma)^{\mathbb{Z}}}$

- objects are orbits in D_Q under action of $(\tau^{-1}\Sigma)^{\mathbb{Z}}$
- morphisms are $\mathrm{Hom}_{\mathcal{C}_Q}(\tilde{X}, \tilde{Y}) = \bigoplus_{p \in \mathbb{Z}} \mathrm{Hom}_{D_Q}(X, (\tau^{-1}\Sigma)^p Y)$
- $\mathcal{C}_Q$ admits a Serre functor $S \cong \tau\Sigma$ induced from that of D_Q
- $\mathcal{C}_Q$ is a 2-Calabi-Yau triangulated category, as $\tau\Sigma \cong \Sigma^2$

Def: a triangulated category is 2-Calabi Yau if it admits a Serre functor $S \cong \Sigma^2$

Def: Serre functor S is unique such that Serre duality holds:

$(\mathrm{Hom}(X,Y))^\vee \Rightarrow \mathrm{Hom}(Y, SX)$ analog of Poincaré duality but for coherent sheaf cohomology

Thm: there is a bijection $\frac{Mesh(\mathbb{Z}Q)}{(\tau^{-1}\Sigma)^{\mathbb{Z}}} \Rightarrow \mathcal{C}_Q$ with cluster structure s.t.

(1) cluster variables of $\mathcal{A}_Q$ are isom classes of indecomposable objects in $\mathcal{C}_Q$

(2) clusters are cluster-tilting sets of indecomposable objects $T_1, \dots, T_n$ such that $\forall i,j$, $Hom(T_i, \Sigma T_j) = 0$

(3) Given cluster-tilting set, the associated endoquiver of the endomorphism algebra of $T = \bigoplus T_i$ has no loops or 2-cycles, and corresponds to the cluster $x_{T_1}, \dots, x_{T_n}$

(4) mutation of cluster-tilting sets through the triangulated category corresponds to quiver mutation

Mutation in $\mathcal{C}_Q$: take a cluster-tilting set $T_1, \dots, T_n$ in $\mathbb{Z}Q$

To mutate at T_k, consider the following exchange triangles

(1) $T_k \xrightarrow{u} \bigoplus_{k \to i} T_i \to T_k^* \to \Sigma T_k$ $\quad$ T_k^* is the cone of $T_k \xrightarrow{u} \bigoplus_{k \to i} T_i$

u is the sum of all the morphisms $T_k \to T_i$

(2) $\bigoplus_{j \to k} T_j \xrightarrow{v} T_k$ has corresponding $\Sigma^{-1} \bigoplus_{j \to k} T_j \xrightarrow{\Sigma^{-1} v} \Sigma^{-1} T_k \to {}^*T_k \to \bigoplus_{j \to k} T_j$

which has a cone ${}^*T_k \cong T_k^*$

shifting twice gives ${}^*T_k \to \bigoplus_{j \to k} T_j \xrightarrow{v} T_k \to \Sigma {}^*T_k$ $\quad$ exact Δ

Then, $T_k^* \cong {}^*T_k$ and this gives exchange relation

$$x_{T_k} x_{T_k^*} = x_{\bigoplus_{j \to k} T_j} \cdot x_{\bigoplus_{k \to i} T_i} = \prod_{j \to k} x_{T_j} \cdot \prod_{k \to i} x_{T_i}$$